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Q1. ATM Withdrawal – Exception Handling Debugging

CODE

```
import java.util.Scanner;
import java.util.InputMismatchException;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        try{
            System.out.print("Enter account balance: ");
            int balance=sc.nextInt();
            System.out.print("Enter withdrawal amount: ");
            int amount=sc.nextInt();
            if(amount<0){
                System.out.println("Transaction failed:Invalid amount");
                return;
            }
            if(amount==0){
                System.out.println("Transaction failed: Cannot divide by zero");
                return;
            }
            if(amount>balance){
                System.out.println("Insufficient balance");
            }else{
                int remaining=balance/amount;
                System.out.println("Withdrawal successful");
                System.out.println("Remaining balance= "+(balance-amount));
            }
        }catch(InputMismatchException e){
            System.out.println("Transaction failed: Invalid input type");
        }catch(ArithmeticException e){
            System.out.println("Transaction failed: Arithmetic error");
        }finally{
            System.out.println("Transaction completed");
        }
    }
}
```

INPUT
500
600

OUTPUT
Enter account balance: Enter withdrawal amount: Insufficient balance
Transaction completed



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SIMATS Engineering • Department of CSE • CSA0905 — Programming In java • 02-09-2026 • Category: CO3-AT3-Debugging and Optimization • Student Submissions Report

Q2. Hospital Registration – NullPointerException Debugging

CODE

```
import java.util.Scanner;
import java.util.InputMismatchException;
class PatientRegistration{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        try{
            System.out.print("Enter patient name: ");
            String inputName=sc.nextLine();
            String name;
            if(inputName==null||inputName.trim().equalsIgnoreCase("NULL")||inputName.trim().isEmpty()){
                name=null;
            }else{
                name=inputName;
            }
            System.out.print("Enter age: ");
            int age=sc.nextInt();
            if(age<0){
                System.out.println("Registration failed: Invalid age input");
                return;
            }
            if(name!=null){
                System.out.println("Patient Name: "+name.toUpperCase());
            }else{
                System.out.println("Patient Name: UNKNOWN");
            }
            if(age>=18){
                System.out.println("Adult patient");
            }else{
                System.out.println("Minor patient");
            }
            System.out.println("Registration completed");
        }catch(InputMismatchException e){
            System.out.println("Registration failed: Invalid age input");
        }finally{
            sc.close();
        }
    }
}
```

INPUT

doe
twenty

OUTPUT

Enter patient name: Enter age: Registration failed: Invalid age input



3. E-Commerce Product Search – Array Exception Debugging

ODE

```
import java.util.Scanner;
import java.util.InputMismatchException;
class ProductSearch{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String[] products={
            "Laptop",
            "Mobile",
            "Tablet",
            "Headphone",
            "Keyboard"
        };
        try{
            System.out.print("Enter product number(1-5): ");
            int productNumber=sc.nextInt();
            if(productNumber<1||productNumber>5){
                System.out.println("Invalid product number ! Please enter a number between 1 and 5.");
            }else{
                System.out.println("Selected product: "+products[productNumber-1]);
            }
        }catch(InputMismatchException e){
            System.out.println("Error: Invalid input type.Please enter an integer.");
        }finally{
            System.out.println("Search completed");
            sc.close();
        }
    }
}
```

INPUT abc

UTPUT Enter product number(1-5): Error: Invalid input type.Please enter an integer.
 Search completed

Q4. Railway Reservation – Synchronization Debugging

CODE

```
class Reservation{
    int seats=1;
    synchronized void bookSeat(String customer){
        if(seats>0){
            System.out.println(customer+"is booking a seat");
            try{
                Thread.sleep(100);
            }catch(InterruptedException e){
                System.out.println("Thread interrupted");
            }
            seats--;
            System.out.println(customer+"booking successful");
        }else{
            System.out.println(customer+"-No seat available");
        }
    }
    public static void main(String[] args){
        Reservation reservation=new Reservation();
        Customer c1=new Customer(reservation,"Customer 1");
        Customer c2=new Customer(reservation,"Customer 2");
        c1.start();
        c2.start();
        try{
            c1.join();
            c2.join();
        }catch(InterruptedException e){
            System.out.println("Main thread interrupted");
        }
    }
}
class Customer extends Thread{
    Reservation reservation;
    String customerName;
    Customer(Reservation reservation,String customer){
        this.reservation=reservation;
        this.customerName=customerName;
    }
    public void run(){
        reservation.bookSeat(customerName);
    }
}
```

OUTPUT

```
Compilation Error:
C:\Windows\TEMP\sandbox_192311391RA_6a860901e48d45.44724230\Reservation.java:39: error: cannot find symbol
        reservation.bookSeat(customerName);
                        ^
    symbol:   variable customerName
    location: class Customer
1 error
```

Q5. Bank Account – Race Condition Debugging

CODE

```
class BankAccount{
    int balance=10000;
    synchronized void withdraw(int amount){
        if(balance>=amount){
            System.out.println(Thread.currentThread().getName()+"withdrawing"+amount);
        }
    }
}
```

